Chapter 4
Limits and Continuity

Note 1

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4.1 : Two-Sided Limits

In the preceding chapter, we studied limits of real sequences. In this chapter we examine the limits of real functions; that is, functions whose domains and ranges are subsets of $\mathbb{R}$.

Recall that a function $f(x)$ converges to a limit L as x approaches a, if $f(x)$ is near L when x is near a. Here is a precise definition of this concept.

Definition 4.1.1 : Let $a \in \mathbb{R}$, let I be an open interval which contains a, and let f be a real function defined everywhere on I except possibly at a. Then $f(x)$ is said to *converge* to L, as x *approaches* a, if and only if for every $\varepsilon > 0$ there is a $\delta > 0$ (which in general depends on ε, f, I, and a) such that

$$0 < |x - a| < \delta \quad \text{implies} \quad |f(x) - L| < \varepsilon \tag{1}$$

or equivalently

$$|f(x) - L| < \varepsilon \quad \text{whenever} \quad 0 < |x - a| < \delta \ . \tag{1'}$$

In this case, we write $L = \lim_{x \to a} f(x)$ or $f(x) \to L$ as $x \to a$, and call L the *limit* of $f(x)$ as x approaches a.

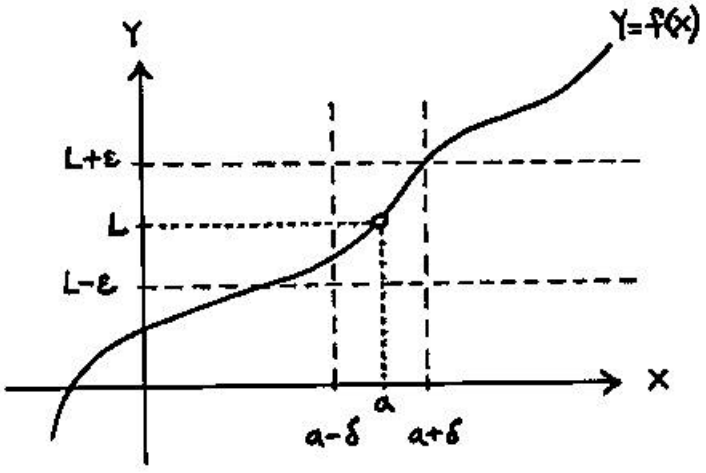


Figure 4.1.1

As was the case for sequences, ε represents the **maximal error** allowed in the approximation $f(x)$ to L. In practice, the number δ represents the **tolerance** allowed in the measurement x of a which will produce an approximation $f(x)$ which is acceptably close to the value L.

According to Definition 4.1.1, to show that a function has a limit, we must begin with a general $\varepsilon > 0$ and describe how to choose a δ which satisfies (1).

Example 1 Suppose that $f(x) = mx + b$, where $m, b \in \mathbb{R}$. Prove that $f(a) = \lim_{x \to a} f(x)$ for all $a \in \mathbb{R}$.

Proof. If $m = 0$, the result holds trivially. Otherwise, given $\varepsilon > 0$, set $\delta = \varepsilon/|m|$. If $|x - a| < \delta$, then

$$\begin{aligned} |f(x) - f(a)| &= |(mx + b) - (ma + b)| \\ &= |m||x - a| \\ &< |m|\delta = \varepsilon \ . \end{aligned}$$

Thus by definition, $f(x) \to f(a)$ as $x \to a$. □

Sometimes, in order to determine δ, we must break $f(x) - L$ into two factors, replacing the less important factor by an upper bound.

Example 2 If $f(x) = x^2 + x - 3$, prove that $f(x) \to -1$ as $x \to 1$.

Proof.

- Let $\varepsilon > 0$ and set $L = -1$. Note that

$$f(x) - L = x^2 + x - 2 = (x - 1)(x + 2) \ .$$

- If $0 < \delta \leq 1$, then $|x - 1| < \delta$ implies $0 < x < 2$.

- So by the triangle inequality, $|x + 2| \leq |x| + 2 < 4$. Set $\delta = \min\{1, \varepsilon/4\}$. It follows that if $|x - 1| < \delta$, then

$$|f(x) - L| = |x - 1||x + 2| < 4|x - 1| < 4\delta \leq \varepsilon \ .$$

- Thus by definition, $f(x) \to L$ as $x \to 1$.

□

The next result shows that even when a function f is defined at a, the value of the limit of f at a is, in general, independent of the value $f(a)$.

Remark : Let $a \in \mathbb{R}$, let I be an open interval which contains a, and let f, g be functions defined everywhere on I except possibly at a. If $f(x) = g(x)$ for all $x \in I \backslash \{a\}$ and $f(x) \to L$ as $x \to a$, then $g(x)$ also has a limit as $x \to a$, and

$$\lim_{x \to a} g(x) = \lim_{x \to a} f(x) \ .$$

Proof.

- Let $\varepsilon > 0$ and choose $\delta > 0$ small enough so that (1) holds and $|x - a| < \delta$ implies $x \in I$.
- Suppose that $0 < |x - a| < \delta$.
- We have $f(x) = g(x)$ by hypothesis and $|f(x) - L| < \varepsilon$ by (1). It follows that $|g(x) - L| < \varepsilon$.

□

Example 3 Prove that $g(x) = \dfrac{x^3 + x^2 - x - 1}{x^2 - 1}$ has a limit as $x \to 1$.

Proof. Set $f(x) = x + 1$ and observe by Example 2 that $f(x) \to 2$ as $x \to 1$ since

$$g(x) = \frac{x^3 + x^2 - x - 1}{x^2 - 1} = \frac{(x + 1)(x^2 - 1)}{x^2 - 1} = f(x)$$

for $x \neq \pm 1$, it follows from the Remark that $g(x)$ has a limit as $x \to 1$ (and that limit is 2). □

There is a close connection between limits of functions and limits of sequences.

Theorem 4.1.1 : (**Sequential Characterization of Limits**)

Let $a \in \mathbb{R}$, let I be an open interval which contains a, and let f be defined everywhere on I except possibly at a. Then

$$L = \lim_{x \to a} f(x)$$

exists if and only if $f(x_n) \to L$ as $n \to \infty$ for every sequence $\{x_n\} \in I \backslash \{a\}$ which converges to a as $n \to \infty$.

Proof.

"$\Longrightarrow$":

- Suppose that f converges to L as x approaches a. Then given $\varepsilon > 0$ there is a $\delta > 0$ such that (1) holds.
- If $\{x_n\} \in I\backslash\{a\}$ converges to a as $n \to \infty$, then choose an $N \in \mathbb{Z}^+$ such that $n \geq N$ implies $|x_n - a| < \delta$.
- Since $x_n \neq a$, it follows from (1) that $|f(x_n) - L| < \varepsilon$ for all $n \geq N$. Therefore, $f(x_n) \to L$ as $n \to \infty$.

"$\Longleftarrow$":

- Suppose that $f(x_n) \to L$ as $n \to \infty$ for every sequence $\{x_n\} \in I\backslash\{a\}$ which converges to a.
- If f does not converge to L as x approaches a, then there is an $\varepsilon > 0$ (call it ε_0) such that the implication "$0 < |x - a| < \delta$ implies $|f(x) - L| < \varepsilon_0$" does not hold for any $\delta > 0$.
- Thus, for each $\delta = 1/n$, $n \in \mathbb{Z}^+$, there is a point $\{x_n\} \in I$ which satisfies two conditions: $\boxed{0 < |x_n - a| < 1/n}$ and $\boxed{|f(x_n) - L| \geq \varepsilon_0}$.
- Now the first condition and the **Squeeze Theorem** imply that $x_n \neq a$ and $x_n \to a$ so by the hypothesis, $f(x_n) \to L$, as $n \to \infty$. In particular, $|f(x_n) - L| < \varepsilon$ for n large, which **contradicts** the second condition.

□

Thus to show that the limit of a function f does not exist as $x \to a$, we need only find two sequences converging to a whose images under f have different limits.

Example 4 Prove that

$$f(x) = \begin{cases} \sin\dfrac{1}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

has no limit as $x \to 0$.

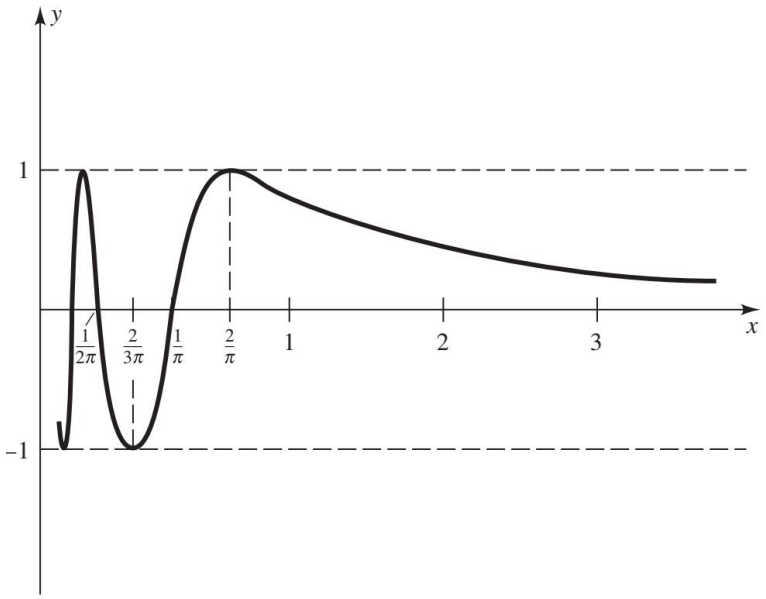


Figure 5.1.2

Proof. By examining the graph of $y = f(x)$, we are led to consider two extremes:

$$a_n := \frac{2}{(4n+1)\pi} \quad \text{and} \quad b_n := \frac{2}{(4n+3)\pi}, \quad n \in \mathbb{Z}^+.$$

Clearly, both a_n and b_n converge to 0 as $n \to \infty$. On the other hand, since $f(a_n) = 1$ and $f(b_n) = -1$ for all $n \in \mathbb{Z}^+$, $f(a_n) \to 1$ and $f(b_n) \to -1$ as $n \to \infty$. Thus by **Theorem 5.1.1** the limit of $f(x)$, as $x \to 0$, cannot exist. □

Theorem 4.1.1 also allows us to translate results about limits of sequences to results about limits of functions. The next three theorems illustrate this principle.

Before stating these results, we introduce an algebra of functions. Suppose that $f, g : E \to \mathbb{R}$. For each $x \in E$, the *"pointwise sum"*, $f + g$, of f and g is defined by $(f + g)(x) := f(x) + g(x)$,

the *"scalar product"*, αf, of a scalar $\alpha \in \mathbb{R}$ with f, by $(\alpha f)(x) := \alpha f(x)$,

the *"pointwise product"*, fg, of f and g, by $(fg)(x) := f(x)g(x)$,

and (when $g(x) \neq 0$) the *"pointwise quotient"*, f/g, of f and g, by

$$\left(\frac{f}{g}\right)(x) := \frac{f(x)}{g(x)} .$$

The following result is a function analogue of **Theorem 3.2.4**.

Theorem 4.1.2 : Suppose that $a \in \mathbb{R}$, that I is an open interval which contains a, and that f, g are defined everywhere on I except possibly at a. If $f(x)$ and $g(x)$ converge as x approaches a, then so do $(f+g)(x), (fg)(x), (\alpha f)(x)$ and $(f/g)(x)$ (when the limit of $g(x)$ is non-zero). In fact,

$$\lim_{x\to a}(f+g)(x) = \lim_{x\to a} f(x) + \lim_{x\to a} g(x)$$

$$\lim_{x\to a}(\alpha f)(x) = \alpha \lim_{x\to a} f(x),$$

$$\lim_{x\to a}(fg)(x) = \lim_{x\to a} f(x) \lim_{x\to a} g(x),$$

and (when the limit of $g(x)$ is non-zero)

$$\lim_{x\to a}\left(\frac{f}{g}\right)(x) = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)} \ .$$

Proof of Theorem 4.1.2.

- Let
$$L \coloneqq \lim_{x\to a} f(x) \quad \text{and} \quad M \coloneqq \lim_{x\to a} g(x) \ .$$

- If $\{x_n\} \in I\backslash\{a\}$ converges to a, then by **Theorem 4.1.1**, $f(x_n) \to L$ and $g(x_n) \to M$ as $n \to \infty$.

- By **Theorem 3.2.4: (i)**, $f(x_n) + g(x_n) \to L + M$ as $n \to \infty$. Since this holds for any sequence $\{x_n\} \in I\backslash\{a\}$ which converges to a, we conclude by **Theorem 5.1.1** that
$$\lim_{x\to a}(f+g)(x) = \lim_{x\to a} f(x) + \lim_{x\to a} g(x) \ .$$

- The other rules follow in an analogous way from **Theorem 3.2.4 (ii)** through **(iv)**.

□

Similarly, the Sequential Characterization of limits can be combined with the Squeeze and Comparison Theorems for sequences to establish the following results.

Theorem 4.1.3 : (**Squeeze Theorem for Functions**)

Suppose that $a \in \mathbb{R}$, that I is an open interval which contains a, and that f, g, h are defined everywhere on I except possibly at a.

(i). If $g(x) \leq h(x) \leq f(x)$ for all $x \in I \backslash \{a\}$, and

$$\lim_{x \to a} f(x) = \lim_{x \to a} g(x) = L \ ,$$

then the limit of $h(x)$ exists, as $x \to a$, and

$$\lim_{x \to a} h(x) = L \ .$$

(ii). If $|g(x)| \leq M$ for all $x \in I \backslash \{a\}$ and $f(x) \to 0$ as $x \to a$, then

$$\lim_{x \to a} f(x)g(x) = 0 \ .$$

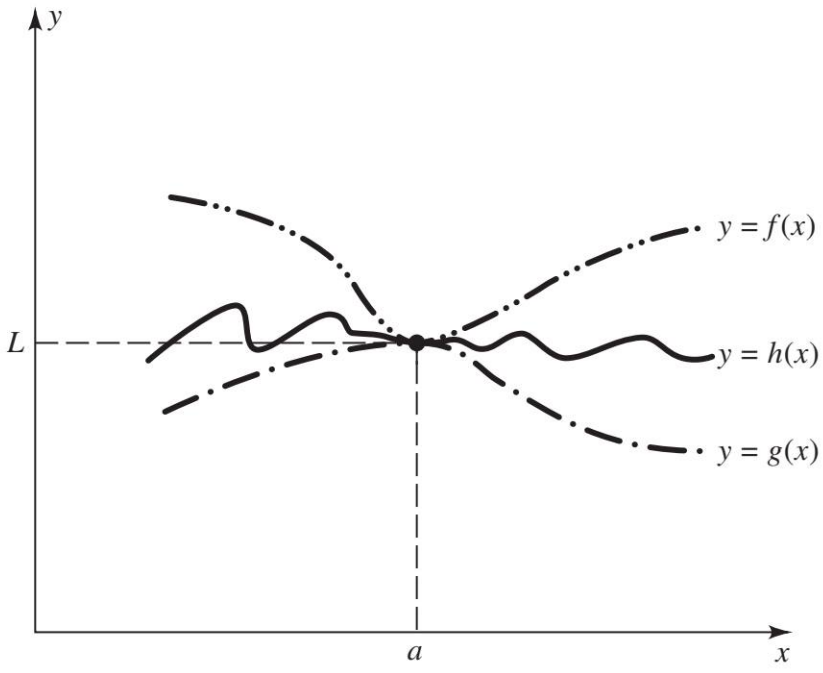


Figure 4.1.3

Theorem 4.1.4 : (**Comparison Theorem for Functions**)

Suppose that $a \in \mathbb{R}$, that I is an open interval which contains a, and that f, g are defined everywhere on I except possibly at a. If f and g have limits as x approaches a and $f(x) \leq g(x)$ for all $x \in I \backslash \{a\}$, then

$$\lim_{x \to a} f(x) \leq \lim_{x \to a} g(x) \ .$$

We shall refer to this last result as taking the limit of an inequality. The limit theorems (**Theorems 4.1.2, 4.1.3,** and **4.1.4**) allow us to prove that limits exist without resorting to ε's and δ's.

Example 5 Prove that $\lim_{x \to 1} \dfrac{x-1}{3x+1} = 0$.

Proof. By Example 1, $x - 1 \to 0$ and $3x + 1 \to 4$ as $x \to 1$. Hence, by **Theorem 4.1.2**,

$$\frac{(x-1)}{(3x+1)} \to \frac{0}{4} = 0 \quad \text{as} \quad x \to 1 \ .$$

□

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EXERCISES

(1). Let $a \in \mathbb{R}$ and let f and g be real functions defined at all points x in some open interval containing a except possibly at $x = a$. Decide which of the following statements are true and which are false. Prove the true ones and give counterexamples for the false ones.

(a) For each $n \in \mathbb{Z}^+$, the function $(x-a)^n \sin\left(f(x)(x-a)^{-n}\right)$ has a limit as $x \to a$.

(b) Suppose that $\{x_n\}$ is a sequence converging to a with $x_n \neq a$. If $f(x_n) \to L$ as $n \to \infty$, then $f(x) \to L$ as $x \to a$.

(c) If f and g are finite valued on the open interval $(a-1, a+1)$ and $f(x) \to 0$ as $x \to a$, then $f(x)g(x) \to 0$ as $x \to a$.

(d) If $\lim\limits_{x\to a} f(x)$ does not exist and $f(x) \leq g(x)$ for all x in some open interval I containing a, then $\lim\limits_{x\to a} g(x)$ doesn't exist either.

(2). Using **Definition 4.1.1** prove that each of the following limits exists.

(a) $\lim\limits_{x\to 2} x^2 + 2x - 5 = 3$

(b) $\lim\limits_{x\to 1} \dfrac{x^2 + x - 2}{x - 1} = 3$

(c) $\lim\limits_{x\to 1} x^3 + 2x + 1 = 4$

(d) $\lim\limits_{x\to 0} x^3 \sin(e^{x^2}) = 0$

(3). Decide which of the following limits exist and which do not. Prove that your answer is correct. (You can use well-known facts about the values of $\tan x, \cos x$, and $\log x$, e.g., that $\log x \to -\infty$ as $x \to 0^+$).

(a) $\lim\limits_{x\to 0} \tan\left(\dfrac{1}{x}\right)$

(b) $\lim\limits_{x\to 0} x \cos\left(\dfrac{x^2+1}{x^3}\right)$

(c) $\lim\limits_{x\to 1} \dfrac{1}{\log x}$

(4). Evaluate the following limits using results from this section.

(a) $\lim\limits_{x\to 1} \dfrac{x^2 + 2x - 3}{x^3 - x}$

(b) $\lim\limits_{x\to 1} \dfrac{x^n - 1}{x - 1}, \quad n \in \mathbb{Z}^+$